

Name _____ Date _____ Period _____

Parent Functions IV: b^x and log base b - Practice

Algebra II | Unit 1 | Lesson 7 Practice | TEK A2.2.A

Objective: I can evaluate and solve exponential and logarithmic expressions, and name their key attributes.

Section 1 — Recall

1. Evaluate: $3^2 =$ _____ 3 to the 0 power = _____ 3^x at $x = -1$: _____
2. Evaluate $\log_2(16)$ and $\log(1000)$.

Section 2 — Apply

3. Solve $3^x = 81$.
 $x =$ _____
4. Solve: log base 5 of $x = 2$.
 $x =$ _____

Section 3 — Challenge

5. A function has domain all reals, range $(0, \infty)$, y-intercept $(0, 1)$, and a horizontal asymptote at $y = 0$. Name the family, then give one specific example function with base 5.

Family: _____ Example: _____

6. Without evaluating exactly, explain why log base 3 of 0 is undefined - what would have to be true, and why it can't be.

Section 4 — Explain It

7. A classmate says: "Since 2^x always increases, log base 2 of x must always decrease, since they're inverses and inverses always do the opposite thing." Is the classmate correct? Use the tables or graphs to explain.

